



CLINICAL PROGRAMMING BOOTCAMP · MODULE 18

Time-to-Event, and the Payoff

Censoring · ADaM validation · one real table

The last day. Two subjects nothing happened to, and why they matter most.

Hands-on: Notebook 18 (ADTTE) and Notebook 19 (from ADaM to a table)

Not whether — how long

ADAE asks

Did this subject have an event?

Y / N

ADTTE asks

How long until they did — and what about the ones who never did?

days + a censoring flag

One row per subject per parameter — EVERY safety subject, event or not.

ABC-01: 8 rows. 6 events. 2 censored. If you built 6 rows, you dropped the censored subjects — and that is the mistake this whole module exists to prevent.

CNSR runs backwards from every other flag in ADaM

CNSR = 0

the event HAPPENED

CNSR = 1

CENSORED — it had not happened when we stopped looking

Everywhere else in ADaM, 1 and 'Y' mean “yes, this is true”. Read it twice, every time.

AND IT APPEARS AGAIN HERE

```
proc lifetest data = adtte;  
  time aval * cnsr(1);      /* <- "1 means censored" */  
  strata trtp;  
run;
```

Write cnsr(0) and every event becomes censored, every censored subject becomes an event. No error. No warning. A curve that is exactly backwards, and a median SAS reports as “not estimable”.

Three ways to answer “how long?” — two are wrong

Mean among subjects
WHO HAD an event

10.17

days $n = 6$

biased DOWN — excludes the long event-free
times precisely because they were long

Mean over EVERYONE,
ignoring censoring

14.62

days $n = 8$

biased UP — asserts two events that never
happened

Kaplan-Meier
median

12

days $n = 8$

uses what each subject actually contributes

“Assume the event happened” is NOT the conservative choice. For time to an adverse event, longer looks safer — so inflating the time flatters the drug. And it fabricates data: a censored subject is one about whom we know one true thing, that they went 28 days event-free. Assigning them an event replaces that truth with a falsehood.

CNSR says we stopped looking. It cannot say why.

LAST DOSE / DATA CUTOFF

Non-informative

the reason has nothing to do with the subject's risk — the KM assumption holds

LAST CONTACT (lost to follow-up)

Possibly INFORMATIVE

people who disappear from studies are sometimes people who felt unwell

With only CNSR, both look identical — two subjects leaving the risk set.

With CNSD TDSC a reviewer can count them separately and judge whether the independent-censoring assumption is credible. That is a question about the VALIDITY OF THE ANALYSIS, not a detail of bookkeeping — which is why studies that censor for more than one reason should carry it. ADaM permits the variable rather than requiring it; recording the reason is a design choice you make, and defend, in the define.

What gets checked, and by whom

Conformance rules

ADaM Conformance Rules v4.0 — structural and metadata rules, machine-checkable. Pinnacle 21 runs these.

Define.xml for analysis data

Every derivation documented, including value-level metadata for each PARAMCD.

Analysis Results Metadata

Links a specific NUMBER in a specific table back to the dataset, the rows, and the code that made it.

Your own assertions

The checks conformance rules cannot know: does AOCCFL count subjects? Is the baseline rule right?

A conformance check confirms the dataset is well-FORMED. It cannot confirm the derivations are RIGHT. That is on you.

THE PAYOFF

Notebook 19 is the shortest in the course.

Two tables that appear in every clinical study report. Two datasets. PROC FREQ and PROC MEANS.
Zero joins. Zero derivations.

which subjects count	SAFFL	ADSL
which events count	TRTEMFL	ADAE
how to count a subject once	AOCCFL / AOCCPFL	ADAE
what “related” means	AREL	ADAE
what order severity prints	ASEVN	ADAE
what counts as baseline	ABLFL	ADVS / ADLB

Every hard decision made ONCE, where it is documented and checked. If they lived in the table programs, two tables in the same submission would eventually disagree — with no error, and no way to tell which was right.

From a number in a table to a page a site filled in



A regulator does not accept a number because a sponsor computed it. They accept it because they can follow it back to a page a site completed, and recompute it at every step. Break one link — a derivation that exists only in a program, a baseline rule that lives in someone's head — and the chain fails there, however right the arithmetic was.

The one deliberate exception: the derived BMI parameter in ADVS has BLANK SRCDOM/SRCSEQ, because it has no single source record. Its traceability runs through define.xml's derivation text instead of through a pointer. A documented link, not a missing one — and being able to explain why is a good final question for this course.